Supplementary Tables

Table 1 presents the set of high-impact data types and their associated Android permissions considered in our study. These include core resources such as calendars, call logs, contacts, location, microphone, sensors, SMS, storage, and others, which represent sensitive access points frequently highlighted in privacy research and regulatory discussions. By explicitly mapping each data type to its relevant permissions, the table provides a concrete scope of the privacy-sensitive behaviors under investigation, forming the foundation for both our system's contextual explanations and the user evaluations of data access practices.

Table 1: The list of our concerned data/resource types and their corresponding permissions.

Data Type	**Permissions**
CALENDAR	READ_CALENDAR WRITE_CALENDAR
CALL_LOG	READ_CALL_LOG WRITE_CALL_LOG
CAMERA	CAMERA
CONTACTS	READ_CONTACTS WRITE_CONTACTS GET_ACCOUNTS
LOCATION	ACCESS_FINE_LOCATION ACCESS_COARSE_LOCATION ACCESS_BACKGROUND_LOCATION
MICROPHONE	RECORD_AUDIO
PHONE	READ_PHONE_STATE READ_PHONE_NUMBERS
SENSORS	BODY_SENSORS BODY_SENSORS_BACKGROUND
ACTIVITY_RECOGNITION	ACTIVITY_RECOGNITION
SMS	SEND_SMS RECEIVE_SMS READ_SMS RECEIVE_WAP_PUSH RECEIVE_MMS READ_CELL_BROADCASTS
STORAGE	READ_EXTERNAL_STORAGE WRITE_EXTERNAL_STORAGE MANAGE_EXTERNAL_STORAGE ACCESS_MEDIA_LOCATION READ_MEDIA_IMAGES READ_MEDIA_VIDEO READ_MEDIA_AUDIO READ_MEDIA_VISUAL_USER_SELECTED

Table 2 presents ten categories of mobile third-party SDKs, grouped according to their primary functionalities. The categories cover a wide spectrum of services, including development utilities, advertising, analytics, mapping, payment processing, social networking, GUI components, game engines, digital identity, and app marketplaces. Each category reflects common forms of third-party integrations in mobile apps, many of which involve sensitive data collection (e.g., advertising and analytics tracking user behavior, digital identity managing authentication data). By organizing third-party services into these functional groups, we provide a clearer view of how external SDKs contribute to app functionality while also shaping privacy risks, offering a systematic basis for analyzing third-party data access and informing user-facing privacy explanations.

Table 2: Types of mobile third-party SDKs

SDK Type	**Description**
Development Aid	Libraries that provide development utilities such as parsers, debugging tools, and code analysis utilities.
Advertisement	SDKs used for integrating in-app advertisements, enabling revenue generation through ad impressions and clicks.
Mobile Analytics	SDKs that collect and analyze user data, tracking user engagement, retention, and performance metrics.
Map	SDKs that provide mapping, navigation, and geolocation services to enhance location-based functionalities in applications.
Payment	SDKs that enable in-app payment processing, supporting various payment gateways and digital wallet integrations.
Social Network	SDKs that integrate social media functionalities, such as sharing, authentication, and social graph interactions.
GUI Component	Libraries providing reusable UI elements, such as widgets, animations, and design frameworks for applications.
Game Engine	SDKs that provide a framework for game development, including physics engines, rendering, and asset management.
Digital Identity	SDKs used for authentication and identity verification, supporting OAuth, biometric authentication, and SSO.
App Market	SDKs that facilitate app distribution, updates, and user acquisition through third-party app stores and marketplaces.

Post-Study Survey

1. Background Information

- Prolific ID: ___________ (Required)
- Daily time spent using mobile apps: ______ (Required)

Types of apps you frequently use: (Select at least one)

Social	Shopping
Weather	Business
Education	Maps and Navigation
Music and Audio	Health and Fitness
Others	

2. Mobile Privacy Preferences

A. Sensitivity of Data Types

Which of the following user data types do you consider sensitive? (Select at least one)

Name and Contact Information	Geographic Location
Call Records	App Usage Records
Biometric Data	Health Data
Bank Transaction Data	Photos

Other sensitive data types (if any): ____

B. Awareness of Privacy Rights

Do you carefully read privacy policies when installing apps? (Required)

◯ Often ◯ Sometimes ◯ Rarely

How familiar are you with setting app permissions? (Required)

◯ Very Familiar ◯ Somewhat Familiar ◯ Not Familiar

C. App Behavior Choices

If you discover an app might leak privacy data, what actions would you take? (Select at least one)

Stop using the app
Look for alternative apps
Continue using but restrict permissions
Contact the developer or relevant authorities

Other actions you might take (if any): ____

3. Risk Attitude Test

1. You participated in a lottery. Which option would you choose? (Required)

◯ A. Guaranteed $90 reward
◯ B. 95% chance to win $100, 5% chance to win nothing

2. You participated in a lucky draw. Which option would you choose? (Required)

◯ A. Guaranteed $5 reward

◯ B. 5% chance to win $100, 95% chance to win nothing

3. You need to pay a fee to participate in a game. Which option would you choose? (Required)

◯ A. Pay $90 for sure

◯ B. 95% chance to pay $100, 5% chance to pay nothing

4. You were told you might need to pay a fine. Which option would you choose? (Required)

◯ A. Pay $5 for sure

◯ B. 5% chance to pay $100, 95% chance to pay nothing

4. User Preferences and Attitudes

Which type of app do you prefer? (Required)

◯ Free apps with extensive permissions

◯ Paid apps without personal data collection

What are your thoughts on mobile privacy protection?